EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 18060

October 2025

ICS 29.220.20; 43.120

English Version

Road vehicles - Rechargeable batteries with internal energy storage - Performance and durability of alkali-Ion (Li-Ion, Na-Ion), Pb, NiMH and combined chemistries EV modules and batteries

Véhicules routiers - Batteries rechargeables avec stockage interne d'énergie - Performance et durabilité des modules et batteries alcalins-ion (Li-Ion, Na-Ion), Pb, NiMH et des chimies combinées pour véhicules électriques

Straßenfahrzeuge - Wiederaufladbare Batterien mit internem Speicher - Unmittelbare Leistung von Modulen und Batterien für Elektrofahrzeuge mit Li-Ion, Pb, NiMH und kombinierter Chemie

This European Standard was approved by CEN on 27 July 2025.

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Ref. No. EN 18060:2025 E

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Annex ZA (informative) Relationship between this European Standard and the requirements of Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC aimed to be covered 27

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European foreword

This document (EN 18060:2025) has been prepared by Technical Committee CEN/TC 301 “Road vehicles”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by April 2026, and conflicting national standards shall be withdrawn at the latest by April 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For the relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website. According to the CEN-CENELEC Internal Regulations, the national standards organisations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom

Introduction

The European Commission published the standardization request M/579 to the European standardization organisations as regards to performance and durability requirements for batteries.

Battery systems are an efficient energy storage system for electrically propelled vehicles. The performance requirements for electric road vehicles battery systems are significantly different from those batteries used for consumer electronics or stationary energy storage. The state of charge window that is accessible for the vehicle application is defined by durability and safety requirements.

This document provides specific performance test procedures for EV batteries.

It enables vehicle manufacturers to choose test procedures to evaluate the characteristics of EV battery systems, battery packs, battery modules and battery cells for their specific requirements.

This document specifies test procedures for alkali-Ion (Li-ion, Na-ion), Pb, NiMH and combined chemistries for EV battery systems, battery packs, battery modules and battery cells to obtain performance and durability values for rated capacity, power, internal resistance, and energy round trip efficiency.

1 Scope

This document describes the performance and durability test procedures and calculation methods for alkali-ion (for example Li-ion, Na-ion), Pb, NiMH and combined chemistries EV battery systems, battery packs, battery modules and battery cells. This document considers the most appropriate metric based on application for electrical performance between different models/products on the market. It specifies performance test procedures and calculation methods to obtain performance and durability values for:

— rated capacity (in Ah) and capacity fade (in %);

— power (in W) and power fade (in %);

— internal resistance (in Ω) and internal resistance increase (in %);

— energy round trip efficiency and its fade (in %);

— expected lifetime (number of reference cycles).

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1
battery
device delivering electrical energy generated by direct conversion of chemical energy, having internal or external storage, and consisting of one or more non-rechargeable or rechargeable battery cells, modules or of packs of them, and includes a battery that has been subject to preparation for re-use, preparation for repurposing, repurposing, or remanufacturing

[SOURCE: Regulation (EU) 2023/1542]

3.2
battery cell
basic functional unit in a battery, composed of electrodes, electrolyte, container, terminals and, if applicable, separators, and containing the active materials the reaction of which generates electrical energy

[SOURCE: Regulation (EU) 2023/1542]

3.3
battery control unit
BCU
electronic device that controls, manages, detects or calculates electric and thermal functions of the *battery system* (3.9) and that provides communication between the battery system and other vehicle controllers

[SOURCE: ISO 12405-4:2018, 3.1]

3.4
battery electric vehicle
BEV
vehicle equipped with a powertrain containing exclusively electric machines as propulsion energy converters and exclusively rechargeable electric energy storage systems as propulsion energy storage systems

[SOURCE: UNECE Global technical regulation No. 15 on Worldwide harmonized Light vehicles Test Procedures]

3.5
battery management system
BMS
electronic device that controls or manages the electric and thermal functions of the battery in order to ensure the battery's safety, performance and service life, manages and stores the data for the parameters for determining the battery's state of health and expected lifetime and communicates with the vehicle, light means of transport or appliance in which the battery is incorporated, or with a public or private charging infrastructure

[SOURCE: Regulation (EU) 2023/1542]

3.6
battery manufacturer
any natural or legal person who manufactures a battery or has a battery designed or manufactured, and markets that battery under its own name or trademark or puts it into service for its own purposes

[SOURCE: Regulation (EU) 2023/1542]

3.7
battery module
set of battery cells that are connected together or encapsulated within an outer casing to protect the cells against external impact, and which is meant to be used either alone or in combination with other modules

Note 1 to entry: Battery modules may or may not include parts of the BMS such as sensors and/or electronics.

Note 2 to entry: Battery module encapsulation can also manage internal stress from the cells.

Note 3 to entry: Battery module is a voluntary design feature of an EV battery.

[SOURCE: Regulation (EU) 2023/1542]

3.8
battery pack
set of battery cells or modules that are connected together or encapsulated within an outer casing to form a complete unit which is not meant to be split up or opened by the end-user

[SOURCE: Regulation (EU) 2023/1542]

3.9
battery producer
manufacturer, importer or distributor or other natural or legal person who, irrespective of the selling technique used, including by means of distance contracts, either:

(i) is established in a Member State and manufactures batteries under its own name or trademark, or has batteries designed or manufactured and supplies them for the first time under its own name or trademark, including those incorporated in appliances, light means of transport or other vehicles, within the territory of that Member State;

(ii) is established in a Member State and resells within the territory of that Member State, under its own name or trademark, batteries, including those incorporated in appliances, light means of transport or other vehicles, manufactured by others, on which the name or trademark of those other manufacturers does not appear;

(iii) is established in a Member State and supplies for the first time in that Member State on a professional basis, batteries, including those incorporated in appliances, light means of transport or other vehicles, from another Member State or from a third country; or

(iv) sells batteries, including those incorporated in appliances, light means of transport or other vehicles, by means of distance contract directly to end-users, whether or not they are private households, in a Member State, and is established in another Member State or in a third country

Note 1 to entry: Distance contracts as defined in Article 2, point (7), of Directive 2011/83/EU (definition 46 in EU Battery Regulation.

[SOURCE: Regulation (EU) 2023/1542]

3.10
battery system
energy storage device that includes cells or cell assemblies or *battery pack(s)* (3.8) as well as electrical circuits and electronics

Note 1 to entry: Examples of electronics are the BCU and contactors.

[SOURCE: ISO 12405-4:2018, 3.3]

3.11
capacity fade
decrease over time and upon usage in the amount of charge that a battery can deliver at the rated current, with respect to the original rated capacity

Note 1 to entry: The EU regulation 2023/1542 states in Annex IV rated voltage instead of rated current under reference conditions

Note 2 to entry: The test for capacity fade is defined in 6.3.3.

[SOURCE: Regulation (EU) 2023/1542]

3.12
charge level
ratio of available charged capacity in relation to actual capacity at the time of assessment

Note 1 to entry: Actual capacity means the amount of charge that a fully charged battery can deliver.

3.13
charge rate
amount of electric current at which a cell or battery is charged

Note 1 to entry: The charge current is expressed as the reference current $I_t = \frac{C_r}{n}$ where C_r is the rated capacity declared by the manufacturer and *n* is the time base in hours for which the rated capacity is declared.

3.14
combined battery and capacitor systems
systems combining both battery component and capacitor component which are used to supply electric energy

3.15
combined chemistry cells
cells with more than one electrochemical system inside the same cell to support different types of operating conditions and improve performance characteristics

Note 1 to entry: Combined chemistry cells can also be denoted as hybrid cells.

Note 2 to entry: A composite electrode material is not per definition the same as a combined chemistry cell.

Note 3 to entry: Combined chemistry cells may comprise a battery component and an electrochemical capacitor component.

3.16
combined chemistry modules
modules with a selection of cells with more than one cell chemistry within or between cells

3.17
combined chemistry packs and systems
packs and systems combining different cell chemistries or cell designs or cell assembly configurations within or between modules or cells

3.18
device under test
DUT
battery system, battery pack, battery module or battery cell

[SOURCE: ISO 12405-4:2018, 3.7, modified — "battery module" added]

3.19
electric vehicle battery
EV battery
battery that is specifically designed to provide electric power for traction in hybrid or electric vehicles of category L as provided for in Regulation (EU) No 168/2013, that weighs more than 25 kg, or a battery that is specifically designed to provide electric power for the traction in hybrid or electric vehicles of categories M, N or O as provided in Regulation (EU) 2018/858

[SOURCE: Regulation (EU) 2023/1542]

3.20
energy round trip efficiency
ratio of the net energy delivered by a battery during a discharge test to the total energy required to restore the initial state of charge by a standard charge

Note 1 to entry: Net energy is the useful electrical energy at the terminals of the DUT.

[SOURCE: Regulation (EU) 2023/1542]

3.21
high-energy battery pack and system
battery pack (3.7) and *battery system* (3.9) using cells, which have the numerical ratio between maximum allowed electric power output and electric energy output at a 1C discharge rate at RT lower than 10

Note 1 to entry: Typically, high-energy battery packs and systems are designed for applications in BEVs and OVC-HEVs.

[SOURCE: ISO 12405-4:2018, 3.12]

3.22
high-power battery pack and system
battery pack (3.7) and *battery system* (3.9) using cells, for which the numerical ratio between maximum allowed electric power output and electric energy output at a 1C discharge rate at RT equal to or higher than 10

Note 1 to entry: Typically, high-power battery packs and systems are designed for applications in NOVC-HEVs.

[SOURCE: ISO 12405-4:2018, 3.13]

3.23
internal resistance
opposition to the flow of current within a cell or a battery under reference conditions, that is, the sum of electronic resistance and ionic resistance to the contribution to total effective resistance including inductive/capacitive properties

Note 1 to entry: The internal resistance is expressed in Ω and defined in 6.4.

[SOURCE: Regulation (EU) 2023/1542]

3.24
internal resistance increase
increase over time and upon usage of the internal resistance of a battery, with respect to the original internal resistance

[SOURCE: Regulation (EU) 2023/1542]

3.25
not off-vehicle charging hybrid electric vehicle
NOVC-HEV
hybrid electric vehicle that cannot be charged from an external source

[SOURCE: UNECE Global technical regulation No. 15 on Worldwide harmonized Light vehicles Test Procedures]

3.26
off-vehicle charging hybrid electric vehicle
OVC-HEV
hybrid electric vehicle that can be charged from an external source

[SOURCE: UNECE Global technical regulation No. 15 on Worldwide harmonized Light vehicles Test Procedures]

3.27
operating charge level window
limited portion of the electrochemical charge level window that is available and used in an application, to meet attribute requirements defined by the system design

Note 1 to entry: The operating charge level window is defined by the OEM.

Note 2 to entry: Different electric vehicle architectures, e.g. mHEV, HEV, PHEV and BEV, operate at different charge level windows to optimize vehicle performance and battery durability.

3.28
power
amount of energy that a battery is capable to provide over a given period under reference conditions

Note 1 to entry: The power is expressed in W.

[SOURCE: Regulation (EU) 2023/1542]

3.29
power fade
decrease over time and upon usage in the amount of power that a battery can deliver at the rated current under reference conditions

Note 1 to entry: The EU regulation states rated voltage instead of rated current under reference condition.

Note 2 to entry: The test for power fade is defined in 6.6.

[SOURCE: Regulation (EU) 2023/1542]

3.30
preparation for repurposing
operation, by which a waste battery, or parts thereof, is prepared so that it can be used for a different purpose or application than that for which it was originally designed

[SOURCE: Regulation (EU) 2023/1542]

3.31
rated capacity
total number of ampere-hours (Ah) that can be withdrawn from a fully charged battery under reference conditions

[SOURCE: Regulation (EU) 2023/1542]

3.32
remanufacturing
technical operation on a used battery that includes the disassembly and evaluation of all its battery cells and modules and the use of a certain number of battery cells and modules that are new, used or recovered from waste, or other battery components, to restore the battery capacity to at least 90 % of the original rated capacity, and where the state of health of all individual battery cells does not differ more than 3 % between cells, and results in the battery being used for the same purpose or application than the one for which the battery was originally designed

[SOURCE: Regulation (EU) 2023/1542]

3.33
repurposing
operation that results in a battery, that is not a waste battery, or parts thereof being used for a purpose or application other than that for which the battery was originally designed

[SOURCE: Regulation (EU) 2023/1542]

3.34
re-use
operation by which products or components that are not waste are used again for the same purpose for which they were conceived

[SOURCE: Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives]

3.35
room temperature
RT
temperature of (23 ±5) °C

[SOURCE: ISO 21498-2:2021, 6.1.8]

3.36
status of the battery
information on the historical and actual use of the battery

Note 1 to entry: During the battery life the status of the battery will change and can be assigned one of the following status: 'original', 'repurposed', 're-used', 'remanufactured' or 'waste'.

Note 2 to entry: A battery has the status 'original' before it is assigned another status (re-used, remanufactured etc…) and when the battery is placed on the EU market. When exactly that happens, depends on the supply chain (a battery in an imported vehicle is normally only placed on the EU market once the vehicle is imported, but a battery that is sold within the EU from a battery manufacturer to a vehicle manufacturer is normally placed on the EU market as a result of that sale).

Note 3 to entry: The note 1 is coming from regulation (EU) 2023/1542.

3.37
technical specification
document that prescribes technical requirements to be fulfilled by a product, process or service

[SOURCE: Regulation (EU) 2023/1542]

4 Abbreviated terms

For the purposes of this document, the following abbreviated terms apply:

ACT	actual
BCU	battery control unit
BEV	battery electric vehicle
BMS	battery management system
BOL	beginning of life
DOD	depth of discharge
DUT	device under test
EV	electric vehicle
HEV	hybrid electric vehicle
mHEV	mild hybrid electric vehicle
NOVC-HEV	not off-vehicle charging hybrid electric vehicle
OCV	open circuit voltage
OEM	original equipment manufacturer
OVC-HEV	off-vehicle charging hybrid electric vehicle
PHEV	plug-in hybrid electric vehicle

RT	room temperature
SC	standard cycle
SCH	standard charge
SDCH	standard discharge

5 General requirements

5.1 Prerequisites for performance tests

A DUT to be tested according to this document shall fulfil the following requirements:

— The necessary documentation for operation and part numbers of the interface parts for connection to the test equipment (i.e. connectors, plugs including cooling, communication) shall be delivered together with the DUT.

— The battery manufacturer shall provide a documentation how to run the battery on the test bench for the specified tests, including operating limits and relevant communication signals.

NOTE 1 Country specific safety requirements apply.

The DUT shall comprise all parts defined by the technical specification.

If not otherwise specified, each full charge or full discharge and each targeted change in charge level shall be followed by a rest period of at least 30 min and until the average OCV is within the tolerances in 5.2.

NOTE 2 For some chemistries, longer rest period may be required.

If not otherwise specified, before each test the DUT shall be equilibrated at the test temperature. Thermal equilibration of the DUT is reached (±4 K) inside the DUT for 15 min.

Testing performed at cell level and scaled or validated at the module, pack or system level is accepted only for beginning of life model/product.

The tests shall be performed at RT or different temperature specified by the manufacturer.

5.2 Accuracy of measurement equipment and measured values

For DUT, the accuracy of external measurement equipment shall be at least within the following tolerances:

— voltage ± 0,5 %;

— current ± 0,5 % for current ≤ 1000 A and ± 5 A for current > 1000 A;

— temperature ± 1 K (in a range from 10 °C to 45 °C).

The overall accuracy of externally controlled or measured values shall be at least within the following tolerances:

— voltage ± 3 %;

— current ± 2 % for current ≤ 250 A and ± 5 A for current > 250 A;

— temperature ± 2 K (in a range from 10 °C to 45 °C);

— time ± 0,1 %;

— mass ± 1 %;

— dimensions ± 1 %.

6 Requirements

6.1 Preconditioning cycles

6.1.1 Purpose

The DUT shall be conditioned by performing at least one electrical cycle, before starting the real testing sequence, in order to ensure an adequate stabilization of the DUT.

6.1.2 Test procedure

The test shall be performed at RT or different temperature specified by the manufacturer.

The discharges shall be performed at C/3 or at a different current if recommended by the battery manufacturer in testing at beginning of life until the end conditions defined by the battery manufacturer.

Consecutive preconditioning cycles shall be performed for the cells in the DUT to be balanced before testing as defined by the battery manufacturer.

At the end of discharge, the DUT voltage shall not go below the minimum voltage limit in the technical specification.

A standard charge (SCH) shall be performed as described in 6.2.2.3.

The results of the pre-conditioning can be checked for consistency.

6.2 Standard cycle (SC)

6.2.1 Purpose

The purpose of the SC is to ensure the same initial condition for each test of a DUT.

6.2.2 Test procedure

6.2.2.1 General

The SC shall be performed at RT or different temperature specified by the manufacturer. The SC shall comprise a standard discharge (SDCH), see 6.2.2.2, followed by a standard charge (SCH), see 6.2.2.3.

All values (time, temperature, current and voltage) shall be noted at least in steps of every 1 % of the estimated discharge and charge time, except if it is noted otherwise in the individual test procedure.

NOTE Depending on the technology, it may be necessary to activate the battery thermal management system during the test.

6.2.2.2 Standard discharge (SDCH)

For a DUT, the discharge rate shall be according to the technical specifications.

NOTE 1 Typical discharge rate for Li-Ion and Na-Ion battery technologies, C/3 for high-energy DUT and 1C for high-power DUT.

NOTE 2 For Pb and NiMH DUT the usual discharge rate is C/3.

Discharge limit is according to the technical specifications given by the manufacturer.

During the discharge process outlined above, the DUT shall not be put in a state outside its specified operating limits.

For a system with multiple battery packs the requirement applies to each battery pack individually and not to the full battery system.

A rest period shall be required to reach a stable condition:

The continuously measured OCV during a 15 min period shall not vary more than ± 2 %.

6.2.2.3 Standard charge (SCH)

For a DUT the charge procedure and end of charge criteria shall be according to the technical specifications.

NOTE 1 Typical charge rate for Li-Ion and Na-Ion battery technologies, C/3 for high-energy DUT and 1C for high-power DUT.

NOTE 2 For Pb and NiMH DUT the usual discharge rate is 1C.

The specifications shall cover end-of-charge criteria and time limits for the overall charging procedure.

For a system with multiple battery packs the requirement applies to each battery pack individually and not for the full battery system.

A rest period shall be required to reach a stable condition:

The continuously measured OCV during a 15 min period shall not vary more than ± 2 %.

6.3 Rated capacity (in Ah) and capacity fade (in %)

6.3.1 Purpose

This is a reference capacity intended to be used to monitor the progression of ageing and the associated capacity fade during the entire lifecycle of the battery.

For the DUT, the constant current discharge rate shall correspond to the manufacturer's C-rate for the capacity measurement. The C-rate can differ between a high energy DUT versus a high-power DUT. The discharge shall be terminated on manufacturer specified discharge voltage limits depending on discharge rates and temperature.

For multiple pack systems the measurement protocol applies to each battery pack individually and not to the full battery system.

Simplified requirements for individual battery testing due to change of status (status 're-used' and "remanufactured") are given in Annex A.

6.3.2 Test procedure

The test sequence for the capacity test at RT or different temperature specified by the manufacturer shall be performed as specified below:

— thermal equilibrium (5.1) and preconditioning cycles (6.1);

— first SC (6.2);

— second SC (6.2).

All discharge tests shall be terminated according to technical specification discharge voltage limits.

After discharge, the DUT shall rest at least for 30 min or shall be thermally equilibrated at the requested ambient temperature, or a fixed time period shall be used to allow for thermal equilibration before starting the next step in the test sequence.

6.3.3 Determination of the rated capacity (in Ah) and capacity fade (in %)

During the second SC, evaluate the capacity under the C rate specified in the technical specification.

The capacity of the DUT is obtained by integrating the discharge current over time.

For the expected lifetime report and after any change of status after "original", the capacity fade can be determined as $\frac{(Q_{BOL} - Q)}{Q_{BOL}} x100$ where

Q_{BOL} is the original rated capacity in Ah;

Q is the current capacity in Ah;

6.4 Determination of internal resistance (in Ω) and internal resistance increase (in %)

Ensure a charge level specified by the manufacturer within the operation window of the charge level as defined by the capacity measurement.

NOTE Longer rest time may be required depending on cell chemistry, battery design etc.

For multiple pack systems the measurement protocol applies to each battery pack individually and not to the full battery system.

Simplified requirements for individual battery testing due to change of status (status "re-used" and "remanufactured") are given in Annex A.

The reference constant current, which will be used to determine the internal resistance, shall be C/5 or any higher current specified by manufacturer.

Internal resistance is determined during the discharge power test pulse (see Figure 1 and 6.6) by measuring the voltage of the DUT before discharge pulse and at least 10 s after start of the pulse.

During discharge the current is negative.

The internal resistance of the pack or the module is calculated by Formula (1):

$$R = \frac{U_0 - U_{10}}{I_0 - I_{10}} \quad (1)$$

where

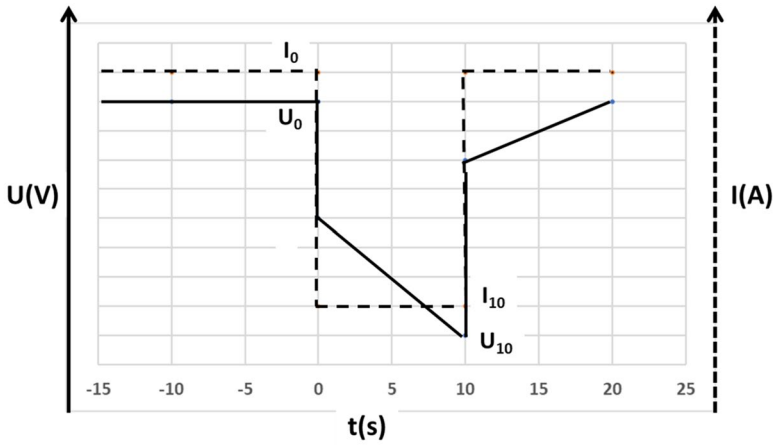
R is the internal resistance of the pack or module in Ω ;

U_0 is the battery voltage of the DUT prior to the pulse in V;

U_{10} is the battery voltage of the DUT during the pulse after 10 s of current in V;

I_0 is the current intensity prior the pulse in A;

I_{10} is the current intensity during the pulse after 10 s in A.



Key

U voltage in V

I intensity in A

t time in s

Figure 1 — Test pulse for power and internal resistance determination

For the expected lifetime report and after any change of status after "original" the internal resistance increase can be determined as $\frac{\left(R - R_{BOL}\right)}{R_{BOL}} x100$

Where

R_{BOL} is the original internal resistance in Ω;

R is the current internal resistance in Ω.

6.5 Testing of partly charged DUT

When a partly charged DUT is to be tested, the DUT shall first be fully discharged under the conditions of a SDCH laid out in 6.2.2.2.

DUT shall then be charged partly under the conditions of a SCH until the needed charging level is reached, unless otherwise specified by manufacturer. This shall be followed by a rest period as specified in 6.2 and until the average OCV is within the tolerances in this document. Then the test shall be performed as described in relevant section of this document.

6.6 Determination of power (in W) and power fade (in %)

Apply the reference current according to the technical specification for the voltage at 20 %, or 50 % or 80 % charge level in the operating window of the application, charge level(s) defined by decision from the manufacturer.

Monitor the voltage for 10 s.

The power is calculated by Formula (2):

$$P = I_{10} \frac{1}{t_{10} - t_9} \int_{t_9}^{t_{10}} U dt \quad (2)$$

where

P is the power in W;

U_9 is the voltage at 9 s;

U_{10} is the voltage at 10 s;

I_{10} is the current at 10 s;

t_9 is the 9th second;

t_{10} is the 10th second.

For a system with multiple packs the measurement protocol applies to each battery pack individually and not to the full battery system.

Simplified requirements for individual battery testing due to change of status (status "re-used" and "remanufactured") are given in Annex A.

Power capability at 20 % or 50 % or 80 % of the charge level have to be reported.

For the expected lifetime report and after any change of status after "original" the power fade can be determined as $\frac{\left(P_{BOL} - P\right)}{P_{BOL}} x100$ where

P_{BOL} is the original power in W;

P is the current power in W.

6.7 Determination of energy round trip efficiency and its fade (in %)

The energy round trip efficiency is determined by a SC.

For multiple pack systems the requirement applies to each battery pack individually and not to the full battery system.

Simplified requirements for individual battery testing due to change of status (status 're-used' and "remanufactured") are given in Annex A.

Energy round trip efficiency (in %) is the ratio of the net DC energy delivered by a DUT during a discharge test to the total DC energy required to restore the initial charge level by a standard charge.

Measure these energies during the last part of the procedure 6.3.3.

η is the energy round trip efficiency (in %), which is the net discharge energy during second standard discharge (6.2.2.2) divided by total charge energy during second standard charge (6.2.2.3) according to Formula (3).

$$\eta = \frac{\int P_{SDCH}\,\mathrm{d}t}{\int P_{SCH}\,\mathrm{d}t} \tag{3}$$

where

P_{SDCH} is the power during the standard discharge of the 2nd standard cycle;

P_{SCH} is the power during the standard charge of the 2nd standard cycle.

Report the net DC discharge energy during SDCH and the calculated η.

For the expected lifetime report and after any change of status after “original” the energy round trip efficiency fade can be determined as $\frac{(\eta_{BOL} - \eta)}{\eta_{BOL}} x100$ where

η_{BOL} is the original energy round trip efficiency in %;

η is the current energy round trip efficiency in %.

6.8 Determination of nominal power

By decision from the manufacturer, nominal power is evaluated from pulses with current at RT at 20 %, or 50 % or 80 % charge level in the operating window at the end of the 10s pulse according to Formula (4).

$$P = I_{10} \frac{1}{t_{10} - t_9} \int_{t_9}^{t_{10}} U \mathrm{d}t \quad (4)$$

NOTE 1 The allowed electric power output is expressed as power in watt (W).

NOTE 2 Typically high-energy battery packs and systems are designed for applications in BEVs and OVC-HEVs.

NOTE 3 Typically high-power battery packs and systems are designed for applications in NOVC-HEVs.

6.9 Energy

The energy is determined during the capacity test (6.3) in the charge level 0 % (CL0) to 100 % (CL100) operating window according to Formula (5).

$$E = \int_{CL0}^{CL100} UI \mathrm{d}t \quad (5)$$

6.10 Ratio of nominal power and discharge energy

The ratio of nominal power (6.8) and discharge energy (6.9) is determined.

NOTE The electric energy output is expressed as energy in watt hours (Wh).

6.11 Expected lifetime

The battery manufacturer defines a reference cycle to determine the expected lifetime, depending on the chemistry, the application and the region of commercialisation.

The severity of this reference cycle shall be specified by declaring the following parameters in the report:

— depth of discharge;

— average charge level;

— operating temperature;

— charge and discharge profiles; and

— rest period between charge and discharge cycles.

In addition, the following shall be determined at the end of the last reference cycle:

— capacity fade;

— power fade;

— internal resistance increase; and

— energy round trip efficiency fade.

For cycle ageing, the expected lifetime shall be expressed in number of full equivalent cycles which is determined by the total discharged and accumulated battery capacity throughout the expected life of the battery divided by the discharged capacity at BOL.

NOTE In the case of a constant DOD standard cycle, the full equivalent cycle number can be estimated by multiplying the number of repetitions with the DOD value.

For calendar ageing, the expected lifetime shall be specified in calendar years, where another reference cycle can be used.

7 Test report

The following data shall be reported:

At original status:

- rated capacity (in Ah) and applied C-rates;
- power (in W);
- power capability at 80 % of the charge level (in W), if possible;
- power capability at 50 % of the charge level (in W), if recommended by manufacturer;
- power capability at 20 % of the charge level (in W), if possible;
- internal resistance (in ω);
- charge level (in %) and reference current (in A) applied for internal resistance measurement;
- energy round trip efficiency (in %);
- ratio between nominal battery power (W) and battery energy (Wh);
- expected lifetime (in number of full equivalent cycles) with the capacity fade, power fade, internal resistance increase and energy efficiency fade at the last full cycle.
- expected lifetime (in calendar years);
- reference cycle used for expected lifetime;
- date of test report.

After any change of battery status (e.g. ‘re-used’, or 'remanufactured') the following data shall also be reported:

- status;
- capacity fade (in %);
- power fade (in %);

- internal resistance increase (in %);
- energy round trip efficiency fade (in %);
- date of test report.

Annex A
(normative)

Simplified requirements for individual battery testing due to change of status (status ‘re-used’, ‘remanufactured’ and ‘preparation for repurposing’)

A.1 Preconditioning – purpose

The DUT shall be conditioned (balancing, charge level and temperature adjustment) based on a defined procedure and target values specified by the manufacturer, before starting the testing sequence, in order to ensure an adequate stabilization of the DUT performance.

This test applies to the DUT.

The DUT shall be preconditioned according to the specification of the manufacturer.

A.2 Capacity (in Ah)

A.2.1 Purpose

For the DUT, the constant current discharge rate shall correspond to the manufacturer’s C-rate for the capacity measurement. The C-rate can differ between a high energy DUT versus a high-power DUT. The discharge shall be terminated on manufacturer specified discharge voltage limits depending on discharge rates and temperature.

For multiple pack system the requirement applies to each battery pack individually and not for the full battery system.

A.2.2 Test procedure

Test sequence capacity test at RT or different temperature specified by the manufacturer shall be performed as specified below:

1. Preconditioning according to A.1, which includes balancing, charge level and temperature adjustment.

2. Run a reference discharge for capacity determination, which can be adjusted by the manufacturer. Reference discharge:

    a. start – charge level: ≥ 80 %;

    b. end – charge level: ≤ 30 %;

    c. C-rate as defined by manufacturer.

All discharge tests shall be terminated within the technical specification discharge voltage limits.

After discharge, the DUT shall rest at least for 30 min or shall be thermally equilibrated at the requested ambient temperature, or a fixed time period shall be used to allow for thermal equilibration before starting the next step in the test sequence.

A.2.3 Determination of capacity (in Ah)

During the reference discharge the capacity under the C rate specified in the technical specification shall be determined by Formula (A.1):

$$C_{battery,actual} = \frac{C_{discharged}}{CL_{start} - CL_{end}} \quad \text{(A.1)}$$

Where

CL_{start} is charge level at beginning of the discharge;

CL_{end} is charge level at end of the discharge;

$C_{discharged}$ (Ah) is the measured capacity between C*L* start (% level of charge) and C*L* end (% level of charge).

After any change of status after "original" the capacity fade shall be determined.

A.3 Determination of internal resistance (in Ω)

After preconditioning, ensure a charge level as specified by the manufacturer and which is equal to the charge level at type testing at begin of life (status 'original').

NOTE Longer rest time may be required depending on cell chemistry, battery design etc.

For multiple pack system the requirement applies to each battery pack individually and not for the full battery system.

The reference current, which will be used to determine the internal resistance, shall be at least C/5 or any higher current specified by manufacturer.

The reference current, which will be used to determine the internal resistance, shall be equal to the reference current at type testing at begin of life (status 'original').

Internal resistance is determined during the discharge power test pulse (see Figure 1) by measuring the pack voltage before discharge pulse and at least 10s after start of pulse.

The internal resistance of the pack or the module is calculated by Formula (A.2):

$$R_{act} = \frac{U_0 - U_{10}}{I_0 - I_{10}} \quad \text{(A.2)}$$

where

R_{act} is the internal resistance of the pack or module in Ω;

U_0 is the battery pack or module voltage prior to the pulse in V;

U_{10} is the battery pack or module voltage during the pulse after 10s of current in V;

I_0 is the current intensity prior the pulse in A;

I_{10} is the current intensity during the pulse after 10 s in A.

See Figure 1, Test pulse for power and internal resistance determination.

A.4 Testing of partly charged DUT

When a partly charged DUT is to be tested, the charge level of the DUTs and temperature shall first be adjusted as specified by the manufacturer, before the battery rests for at least 30 min.

A.5 Determination of power (in W)

Defined temperature RT or different temperature specified by the manufacturer.

Apply the reference current according to the technical specification at the defined reference charge level for the internal resistance measurement.

Monitor the voltage for at least 10 s.

The power is calculated by Formula (A.3):

$$P_{act} = I_{10} \frac{1}{t_{10} - t_9} \int_{t_9}^{t_{10}} U \mathrm{d}t \tag{A.3}$$

where

P_{act} is the actual power in W;

U_9 is the battery pack or module voltage during the pulse after 9 s of current;

U_{10} is the battery pack or module voltage during the pulse after 10 s of current;

I_{10} is the current intensity during the pulse after 10 s.

For multiple pack system the requirement applies to each battery pack individually and not for the full battery system.

The power capability at 80 %, 50 % and 20 % charge level can be calculated based on the results for the power at the defined reference charge levels for the internal resistance measurement. See Formula (A.4) and (A.5)

$$P_{battery,act}\left(CL\right) = U_{10,act}\left(CL\right) \cdot I_{battery,max}\left(CL\right) \tag{A.4}$$

$$U_{10,act}\left(CL\right) = (U_{0,BoL}\left(CL\right) - (U_{0,BoL}\left(CL\right) - U_{10,BoL}\left(CL\right)) \cdot \frac{R_{act}}{R_{BoL}}) \tag{A.5}$$

Where

$P_{battery,act}(CL)$ is the actual power at the considered Charge Level in W;

$U_{0,BoL}(CL)$ is the battery pack or module voltage during the pulse after 0 s of current at beginning of life at the considered Charge Level;

$U_{10,Bol}(CL)$ is the battery pack or module voltage during the pulse after 10 s of current at beginning of life at the considered Charge Level;

$I_{battery,max}(CL)$ is the maximal current intensity at the considered Charge Level;

R_{act} is the actual resistance;

R_{BoL} is the resistance at beginning of life.

A.6 Determination of energy round trip efficiency (in %)

Defined temperature RT or different temperature specified by the manufacturer.

For multiple pack system the requirement applies to each battery pack individually and not for the full battery system.

Based on the determination of the internal resistance, the influence of the ageing on the round trip efficiency can be determined, using Formula (A.6):

$$\eta_{Roundtrip,actual} = 1 - (1 - \eta_{Roundtrip,BoL}) \cdot \frac{R_{actual}}{R_{BoL}} \quad \text{(A.6)}$$

A.7 Determination of estimated remaining lifetime

The battery capacity or battery energy available at the change of status can become the new BOL value. Calculations are made according to 6.11.

Annex ZA
(informative)

Relationship between this European Standard and the requirements of Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC aimed to be covered

This European Standard has been prepared under Commission Implementing Decision C(2021) 8614 final of 7.12.2021 on a standardization request to the European standardization organisations as regards performance, safety and sustainability requirements for batteries (M/579), to provide one voluntary means of conforming to Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC [OJ L 191].

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding requirements of that Regulation, and associated EFTA regulations, if, where applicable, the legal minimum values for those requirements are attained.

Table ZA.1 — Correspondence between this European Standard and Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC [OJ L 191]

Requirement of Regulation (EU) (EU) 2023/1542	Clause(s)/subclause(s) of this EN	Remarks/Notes
Article 10 Paragraph 1 Annex IV, part A, 1.	6.3	Rated capacity and capacity fade
Article 10 Paragraph 1 Annex IV, part A, 2.	6.6	Power and power fade
Article 10 Paragraph 1 Annex IV, part A, 3.	6.4	Internal resistance and internal resistance increase
Article 10 Paragraph 1 Annex IV, part A, 4.	6.7	Energy round trip efficiency and its fade
Article 10 Paragraph 1 Annex IV, part A, 5.	6.11	Expected lifetime
Article 10 Paragraph 1 Annex IV, part B, 1.	6.2	Applied discharge rate and charge rate in the standard cycle
Article 10 Paragraph 1 Annex IV, part B, 2.	6.10	Ratio of nominal battery power and battery energy
Article 10 Paragraph 1 Annex IV, part B, 3.	6.11	Depth of discharge

Article 10 Paragraph 1 Annex IV, part B, 4.	6.6	Power capability at 20 % and 80 % charge level

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the product(s) falling within the scope of this standard.

Bibliography

[1] Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC

[2] ISO 12405-4:2018, *Electrically propelled road vehicles —Test specification for lithium-ion traction battery packs and systems — Part 4: Performance testing*

[3] ISO 21498-2:2021, *Electrically propelled road vehicles — Electrical specifications and tests for voltage class B systems and components — Part 2: Electrical tests for components*

[4] IEC 60050-482:2004, *International Electrotechnical Vocabulary (IEV) - Part 482: Primary and secondary cells and batteries*

[5] UNECE Global technical regulation No. 15 on Worldwide harmonized Light vehicles Test Procedures

[6] Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives